

Lie Groups and Differential Equations

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August 2022

1 Lecture I

1.1 Introduction

Many current introduction to the theory of Lie groups and Lie algebras present it's modern development, discussing topics such as the classification of Lie groups and algebras or the topology of Lie groups. While these are interesting questions once you understand Lie groups, they often fail to answer the question of why someone should care about such an object in the first place. You can always try to come up with some motivation to study a topic, but what better motivation is there than the reason it was discovered in the first place? The goal of these notes is to present a sense of Lie's original reason for studying these objects: trying to understand differential equations. His motivation came from thinking about Galois theory. Galois took a polynomial and associated to it a group which permuted its roots, and in turn, used this to understand the polynomial. Lie wanted to associate a group to a differential equation which would also permute its solutions and use this to understand the differential equation. He was led to considering "continuous groups" as they admit very powerful calculus techniques to test for the invariance. We will work towards this concept and hopefully the presentation will demonstrate how one could naturally be led to the concept of a Lie group. A note: The referenced textbook "Applications of Lie Groups to Differential Equations" by Oliver is excellently written and requires minimal prerequisites. If you find this material interesting or would like a more in-depth explanation of the concepts here, I would highly recommend reading this book.

1.2 What is a Lie Group

In this section, instead of formally defining a Lie group, I want to give a number of examples so that you have a picture in your mind of the types of objects we are working with. The poor one sentence description is that a Lie group is a group that also has a manifold structure on it. The better description is it's a group that can be parameterized (at least locally) by variables taking values in $\mathbb{R}$. Hopefully the following examples illustrate this phenomenon:

- $S^1 = \{e^{i\theta}\}$ (the unit complex numbers with usual multiplication) is parametrized by the one real parameter θ
- $T^n = (S^1)^n$ (the product of n circles with componentwise multiplication) is a group that can be parametrized with the n angles $\theta_1, \dots, \theta_n$.
- $M^{n \times n}$ ($n \times n$ matrices with the operation of addition) forms a group with n^2 parameters being the entries a_{ij} of the matrix.
- $GL(n, \mathbb{R})$ ($n \times n$ invertible matrices with the usual multiplication) forms a group with n^2 parameters of the entries of the matrix - note that these entries cannot take any value since you may end up with 0 determinant, however you can still vary each entry of an invertible matrix independently by a small amount and still keep it invertible which is the reason it has n^2 parameters.

- $O(n, \mathbb{R})$ ($n \times n$ matrices U satisfying $U^T U = I_n$ with usual multiplication) forms a group - in this case the parameters may not be varied freely and the condition $U^T U = I_n$ imposes polynomial constraints on the entries of the matrix, it can be shown that there are $n(n-1)/2$ independent parameters of the matrix that can be varied, and the remaining entries will have their values determined. Try to find such a set of parameters.

The important idea is that around any group element, you should be able to vary the independent parameters by a small amount and still stay in your group. The amount of independent parameters that can be freely varied is called the group's dimension. Another more technical condition is that the group and topological data must be "compatible" and so group multiplications and inversions must be continuous maps on the inputs. Intuitively, if you multiply $ab = c$ and you choose some group element a' which is "close" to a , and compute $c' = a'b$, then you expect c and c' should also be "close" and likewise if we vary the second factor. The same sort of logic works with inversion: if a' is close to a , we expect that $(a')^{-1}$ is "close" to a^{-1} . Verify that all of these make sense in the examples above. It is also important to see some examples of things which aren't Lie groups, because they can't be parametrized by $\mathbb{R}$. These groups include $\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}, \mathbb{Q}, \mathbb{Q}_l, \mathbb{S}_\infty$ etc. Notice that with these non-Lie groups, we have a sense that they are "discrete" in some way and that Lie groups on the other hand are "continuous." This is the main takeaway of these examples.

Notice that I never defined what a manifold is. There are a few reasons for this. One big reason is that the formal definition is a bit too cumbersome for our purposes and doesn't help with our intuition. For us, it is enough to think of a manifold roughly as "something that can be parametrized with real parameters." Perhaps a stronger reason is that Lie most certainly didn't work with the formal definition of a manifold, partly because the notion was still under development. In fact, he didn't work with Lie groups themselves at all, and instead focused on "Lie group actions." Similar to how our first perception of groups is that of group actions, our first perception of Lie groups will be as some action and we generally won't even touch an abstract Lie group which isn't an action for the remainder of these notes. This action will be some way that the group changes a geometric space in a continuous manner. Here are some examples:

- $\mathbb{R}$ acts on $\mathbb{R}^n$ by translation (along a fixed direction)
- $\mathbb{R}^{>0}$ with the operation of multiplication acts on $\mathbb{R}^n$ by scaling each coefficient (henceforth, we denote this group as $\mathbb{R}^\times$, but we will ignore the negative numbers and only focus on the connected component to the identity)
- S^1 acts on $\mathbb{R}^2$ by rotating the plane
- The action above can also be thought of as an action of $\mathbb{R}$ on $\mathbb{R}^2$ where the parameter of $\mathbb{R}$ represents the angle instead of S^1
- More generally $O(n, \mathbb{R})$ acts as rotations of $\mathbb{R}^n$, and $GL(n, \mathbb{R})$ acts as linear transformations on $\mathbb{R}^n$

I won't state the formal axioms of an action, but you should be able to get the sense of what it is from the examples above, and you should also be able to verify that each example satisfies the usual group action axioms. Note that each of the examples above have the property that if you take an element of the group close to the identity, the transformed point is fairly close to the initial point. More generally, if you take two nearby elements of the group, the image of a single point under both maps will be close to each other. This is the sort of "continuity" condition that we add when we want to intertwine the group action with the topology of the group (also, show that technically the latter description follows from the former).

The important part of group actions being continuous is that it will allow us to take derivatives of these actions in some sense. For example, we can construct the tangent vectors to the rotation action. It is not quite clear why this should be a useful condition to have, but we will try to motivate that as

the notes progress. Another thing to notice is that S^1 and $\mathbb{R}$ both act on the plane with rotation, but "locally" (close to the identity), S^1 and $\mathbb{R}$ look very similar. Since derivatives are a very local property, we won't really care about the global nature of our group which is acting on a space, but only the local nature. This concept of a "local group action" will be expanded upon later when it becomes relevant.

1.3 An Introductory Example

Here, we outline an example which demonstrates the power of using groups to study differential equations. We will study the differential equation $y_{xx} = 0$ (derivatives will be denoted with subscripts). We already know that the solutions to this differential equation correspond to all lines in the plane. We can look at some group actions on the plane which leave send these solutions to other solutions, or other words, transformations of the plane which send lines to lines. Of the examples of Lie group actions mentioned above, we can see that rotations of the plane, translations in any direction, and scaling the plane will all be actions that will send solutions to solutions. Now how can we use this group to understand the differential equation? Suppose we do not know the solutions to the differential equation, but we know any of these transformations of the plane will send the graph of some solution to the graph of another solution. Let's say that after analyzing our equation, we find that $y = 0$ is a very simple solution to our differential equation. Given that rotations send solutions to solutions, we must also have that $y = mx$ is a solution to our differential equation since given some m , we can always rotate the plane to send the graph of $y = 0$ to the graph of $y = mx$. Now, by similar reasoning, we can get any equation of the form $y = mx + b$ since starting from $y = mx$, which we know must be a solution, we can always translate vertically to get $y = mx + b$. Thus, using knowledge of our group of symmetries, we were able to deduce all possible solutions to our differential equation. Of course, this is a particularly special example, and this is behavior that would not occur in general, but at least it outlines why the group of symmetries of a differential equation should give some insight into the equation itself.

Now, you may not be very impressed with this as we determined that the group leaves the solutions invariant because we already knew the solutions, and we used this to derive the solutions. However, we can determine whether a group sends solutions to a differential equation to other solutions just by analyzing the differential equation itself. Here is an example: suppose we have the differential equation $yy_{xx} = (y_x)^2$. Let's analyze what happens when we transform the plane by the action of $\mathbb{R}^\times$ which sends the coordinate $(x, y) \rightarrow (z, u) = (x, y^\alpha)$ for $\alpha \in \mathbb{R}^\times$ (we relabel the coordinates as z, u). This will take the graph of some $y = f(x)$ to $u = (f(z))^\alpha = g(z)$. What it means to say that this group action leaves the differential equation invariant is to say that if $ff'' = (f')^2$, then $gg'' = (g')^2$, where the derivatives are with respect to x in the first equation and z in the second. Let's go through the calculations:

$$u_z = \frac{\partial u}{\partial z} = \frac{\partial y^\alpha}{\partial x} \frac{\partial x}{\partial z} = \alpha y^{\alpha-1} y_x u_{zz} = \frac{\partial \alpha y^{\alpha-1} y_x}{\partial z} = \frac{\partial \alpha y^{\alpha-1} y_x}{\partial x} = (\alpha - 1) \alpha y^{\alpha-2} (y_x)^2 + \alpha y^{\alpha-1} y_{xx}$$

However, we know that $y_{xx} = (y_x)^2/y$ because we're assuming y is the graph of some function that satisfies our differential equation. Hence, we get

$$u u_{zz} = (\alpha - 1) \alpha y^{\alpha-2} (y_x)^2 + \alpha y^{\alpha-1} y_{xx} = (\alpha - 1) \alpha y^{\alpha-2} (y_x)^2 + \alpha y^{\alpha-2} y_x^2 = \alpha^2 (y^{\alpha-2}) y_x^2 = u_z^2$$

Hence, we see that if $y(x)$ satisfies our differential equation, so does our transformed equation $u(z)$. The most trivial solution is $y = 0$, but this doesn't really yield any interesting results. A less trivial solution could come by noticing that the equation is "homogenous" and so if we have $y_x = y$, the equation would be solved. Hence, we can take $y = e^x$ as a solution. Then, our calculation above tells us that $y = e^{\alpha x}$ (after changing (z, u) to (x, y)) is also a solution, which you can easily verify.

We can also show in a similar fashion that the actions that leave $y_{xx} = 0$ invariant without knowing the solutions of the differential equation beforehand. Before we do so, it will be helpful to consider some new language to facilitate our discussion and move us towards the concept of a Lie group. As

you can see from the above example, once we determined the derivatives of u with respect to the derivatives of y , the problem of showing that the action leaves the differential equation invariant was reduced to algebra. This motivates us to view derivatives as geometric variables of their own. This will turn our differential equation into an algebraic equation and make it easier for us to talk about whether a group leaves a differential equation invariant.

1.4 Jet Space

The jet space is a geometric space where our geometric variables are not just the independent and dependent variables, which will be denoted x_i and u going forward, but also the derivatives of the independent variables such as $u_{x_1}, u_{x_1x_1}, u_{x_1x_2}$ etc. This allows us to speak about differential equations and group actions very algebraically. You will notice from the example that the computation for the n -th derivative of the transformed dependent variable depended on no more than at worst the n -th derivatives of the old dependent variables (and possibly the independent variables in general. This is just an exercise in manipulations of Taylor series. Moreover, we see that the algebraic equations that give the derivatives of the transformed function depend continuously on the initial data of the values and derivatives.

Let's see the implications of these observations and outline our plan going forward. We will construct a space called the n -th jet space, whose coordinates consist of all independent and dependent variables. The intuition for this space is it is a space representing all possible functions. Each point in the jet space corresponds to a function which passes through the point (in the base space) prescribed by the independent and dependent variable coordinates, and which has derivatives at that (base space) point prescribed by the derivative coordinates at the point. It may be better to think of this space as all possible n -th order Taylor polynomials since we are ignoring all higher order information and thus cannot distinguish between two functions that are the same up to n -th order. As we mentioned, this is sufficient for n -th order partial differential equations since we won't ever find a term that has higher order derivatives. In this space, we now interpret a differential equation like $u_{xx} = 0$ as an algebraic equation rather than as a differential equation. This particular example corresponds to a hyperplane. Our Lie group action on the base space will then extend to a Lie group action on the jet space. Intuitively, if the points in a space vary continuously, then the derivatives of a graph in this space will also vary smoothly. Algebraically, this will lead us to transformation equations for u_x, u_{xx} etc. in terms of the geometric transformation of (x_i, u) , as we had seen before. This will be called the prolonged group action, and had already been seen before when we found u_x, u_{xx} in terms of y_x, y_{xx} which depended on the parameter α from the group. The statement that the group action sends solutions of a differential equation to other solutions can now be stated more geometrically as follows: the prolonged group action should "preserve" the geometric hypersurface cut out by our differential equation. In terms of our problem of $u_{xx} = 0$, we want to send points on this hyperplane to other points, and since each point represents a function, this will correspond to taking one function that satisfies the differential equation to another function that does. Perhaps a clearer geometric picture, take the following analogy: $\mathbb{R}^n$ is analogous to our jet space representing all functions, $x^2 + y^2 + z^2 = 1$ is analogous of our differential equation, the points on this sphere is our functions that satisfy our differential equation, and an action like rotation around the z -axis is analogous to a group action that sends solutions to our differential equation to other solutions. An action like vertical translation in this analogy will not send the sphere to itself and hence, if we start with a solution to our differential equation, we will not (unless we luck out by coincidence) get back another solution.

Definition 1 (Jet Space). The n -th jet space $U^{(k)}$ is the Euclidean space with variables $(x_1, \dots, x_n, u, u_{x_i}, u_{x_i x_j}, \dots)$ where we take all partial derivatives up to order n . Here, we will not take redundant derivatives so we will not have both $u_{x_1 x_2}$ and $u_{x_2 x_1}$. This can be phrased using multi-indices. A multi-index J is a sequence of variables $J = (x_{i_1}, \dots, x_{i_j})$ with i some non-decreasing sequence which can have repetitions, and the size $|J|$ is defined to be the amount of elements j . Hence, our variables consist of x_i, u and variables labeled with u_J with J any multi-index with $|J| \leq k$. Lastly, $U^{(0)}$ consists of the variables (x_i, u) , which we shall refer to as the base space.

For example, if we have 2 independent variables x, t , our second jet space $U^{(2)}$ has coordinates $(x, t, u_x, u_t, u_{xx}, u_{xt}, u_{tt})$. These variables can take arbitrary values and the coordinate names should just be thought of as labels. In particular, do not think of it as " u_x is the derivative of u ," but rather as " u_x and u are coordinates independent from one another." We will use the interpretation of these variables as derivatives, so it is important to be able to switch our perspectives on these variables when convenient. However, as mentioned, the point $(x_i, u, u_{x_i}, u_{x_i x_j}, \dots)$ in this space correspond to functions which pass through (x_i, u) in the base space and have derivatives $u_{x_i}, u_{x_i x_j}$ etc.

Exercise (Dimension of the Jet Space)

What is the dimension of the k -th jet space if there are n independent variables?

1.5 Returning to the Example

Now that we have developed some terminology, let's show that the action of rotating the plane leaves the differential equation $u_{xx} = 0$ invariant. Since we have second derivatives in our equation, we will want to analyze the second jet space $U^{(2)} = \{(x, u, u_x, u_{xx})\}$. Specifically, we want to see what the group action $\varphi_t(x, u) = (x \cos t - u \sin t, u \cos t + x \sin t)$ extends to the derivatives. In other words, if we have the graph of some function, which has first and second derivatives γ, δ at the point (α, β) , then we want to rotate by some angle t and look at what the first and second derivatives of the resulting graph is at the point $\varphi_t(\alpha, \beta)$. We will then show that if $\delta = 0$, or in other words, our function at some point had an initial second derivative of 0, then the resulting graph will have a second derivative of 0 at the rotated point as well. Note: this is stronger than saying we have a function whose second derivative is zero at every point. The statement works if the second derivative is 0 at any point, or in other words, rotating a function will send an inflection point of a graph to another inflection point.

The algebra will get quite messy, so I will only compute what the resulting first derivative is, in other words, the action on the first jet space $U^{(1)}$. The computation of the action on the second derivative will be left as an exercise. The first jet space has coordinates (x, u, u_x) . The method for computing this action is as follows: we must first find a representative function which passes through the point (α, β) and has a derivative of γ at that point, then look at what that function gets transformed to, and then finally look at what the derivative of the resulting function is. Since our jet space is blind to higher order derivatives, we can just take a first order Taylor series (a line) which has the desired properties and see how it transforms. Hence, we want to look at what the graph of $u = \beta + \gamma(x - \alpha)$ gets sent to. This may seem very restrictive to make a general statement about all functions, but it turns out to give the right answer regardless of higher derivatives. In the exercise later to compute the action of the second derivative, you will need to use a quadratic which has a matching first derivative at this point, and you will see that the group action derived from that calculation is the same.

Let's begin with the calculation. We will first compute the case for $u = ax + b$ for ease of calculation, and then substitute a, b in terms of α, β, γ at the last step. First, we will denote our transformed variables as $(\tilde{x}, \tilde{u}) = \varphi_t(x, u) = (x \cos t - u \sin t, u \cos t + x \sin t)$. Suppose now that $u = ax + b$. We then have $(\tilde{x}, \tilde{u}) = (x \cos t - (ax + b) \sin t, (ax + b) \cos t + x \sin t)$. We now want to solve for $\tilde{u}$ as a function of $\tilde{x}$. To do this, we solve for x as a function of $\tilde{x}$ and substitute this expression into our expression of $\tilde{u}$ as a function of x . We have

$$x = \frac{\tilde{x} + b \sin t}{\cos t - a \sin t}$$

$$\tilde{u} = x(\sin t + a \cos t) + b \cos t = \frac{\sin t + a \cos t}{\cos t - a \sin t} \tilde{x} + \frac{b}{\cos t - a \sin t}$$

Now, if we substitute $a = \gamma$ and $b = \beta - \alpha\gamma$ so that the initial values are met, we get

$$\tilde{u} = \frac{\sin t + \gamma \cos t}{\cos t - \gamma \sin t} \tilde{x} + \frac{\beta - \gamma\alpha}{\cos t - \gamma \sin t}$$

Now, the derivative of this at the transformed point $(\alpha \cos t - \beta \sin t, \beta \cos t + \alpha \sin t)$ is $\frac{\sin t + \gamma \cos t}{\cos t - \gamma \sin t}$. I encourage you to calculate $\tilde{u}(\alpha \cos t - \beta \sin t)$ (evaluating the function on the transformed x coordinate) and verify that you do indeed get $\beta \cos t + \alpha \sin t$ (the transformed u coordinate). This is an example of how our transformation rules on the k -th derivative only depend on the k -th derivative data, since our output can be thought of as 0-derivative data. Now, our group action φ_t on $U^{(0)}$ extends to an action on $U^{(1)}$ given by $\varphi_t(\alpha, \beta, \gamma) = (\alpha \cos t - \beta \sin t, \beta \cos t + \alpha \sin t, \frac{\sin t + \gamma \cos t}{\cos t - \gamma \sin t})$.

Exercise (Computation of Action on $U^{(2)}$)

Now, we want to find how the group action on our base space changes the second derivative. Our coordinates are (x, u, u_x, u_{xx}) . Find some function which passes through the point (α, β) , has first derivative γ , and second derivative δ . Do the same procedure as above: substitute your function $u(x)$ into $(\tilde{x}, \tilde{u}) = (x \cos t - u \sin t, u \cos t + x \sin t)$, solve for $\tilde{x}$ in terms of x , substitute that into your expression for $\tilde{u}$ in terms of x and hence obtain the transformed function $\tilde{u}(\tilde{x})$. Compute the value, first derivative, and second derivative at the point $\tilde{x} = \alpha \cos t - \beta \sin t$ (the transformed initial point). Verify that the value and first derivative match with what we obtained using the line. Your second derivative should come out to $\frac{\delta}{(\cos t - \gamma \sin t)^3}$.

Hence, we get quite trivially (because 0 divided by anything is 0) that if a graph at some point has a second derivative of 0 at a point, the rotated graph will also have a derivative of 0 at the transformed point. Another way to state this is that it takes the hyperplane $u_{xx} = 0$ to itself. Now, a solution to the differential equation will be represented in the jet space as a collection of points on the hyperplane $u_{xx} = 0$, and we have just shown that all of these points will be sent to other points on this hyperplane. Or finally, rotations take some solution to this differential equation, and return another solution to the differential equation.

As the exercise above shows, this method is quite inefficient. Let's look at the general procedure: a group action sends (x_i, u) to $(\tilde{x}, \tilde{u}) = g(x_i, u)$ and so a function $u = f(x)$ gets sent to $\tilde{u} = g(f(g^{-1}\tilde{x}_i))$. In other words, we act on the output with the group action and act on the input with the inverse of the group action. This is something you are already familiar with: if you stretch a graph vertically by 2, multiply the function by 2, but to stretch it horizontally, you must divide the input by 2. Similarly, to translate vertically, we add a constant but to translate horizontally, we must subtract the constant from the input. We saw this phenomenon when we had to invert the transformation to solve for x in terms of $\tilde{x}$ to substitute into the equation for $\tilde{u}$. We had to solve a quadratic just to get the action on the second variable above. To get the action of the third derivative, we would have to invert a cubic, which is already significantly difficult. In the other direction, if we had multiple input variables, we would have to invert some nonlinear functions of multiple variables, which is also extremely difficult. Moreover, we had to make a seemingly arbitrary choice of a function which matched the value and derivatives prescribed, but our final answer was independent of this choice. This suggests that we should look for a new method which would be simpler and hopefully avoid the need to look at an auxiliary function if we hope to stand a chance against understanding groups leaving an equation invariant. Note something else: nowhere above did we use the fact that our actions were coming from a continuous group, or even that they were coming from a group at all; they could have just been arbitrary transformations existing void of anything else and just happen to leave the equation invariant (e.g. fix some value of t and study only that rotation). In what follows, the criterion for wanting some sort of continuity will become clear.

2 Lecture II

2.1 A Better Test for Invariance

Let's go back to our analog of the sphere in $\mathbb{R}^3$ as modeling the solution of our differential equation in the space of all functions. We saw that the group of rotations around the z -axis will leave the sphere invariant. But write down the transformation and show that it leaves the sphere invariant. Take some

other equations: $z = 0$ and $x = 0$. Although these are simple equations, the rotation leaves the first invariant while it doesn't leave the second invariant. If you try to find a generic method to see if an equation is invariant under this transformation, you will find it somewhat difficult. Even if you do find a general method, try to generalize it to other actions like scaling $\mathbb{R}^3$. We can also ask the reverse question of what are all transformations which leave the sphere invariant. For example, we know that if we rotate around some vector v by some angle about the origin, it will still leave the sphere invariant. But do you know the formula for this transformation off the top of your head, and a nice way to show that any such transformation leaves the sphere invariant?

The nice way to combine all of these into one general formulation: look locally. We can take some group action like rotations and look at the tangent vector to this action at each point. What that means is that we look at what path a point would trace out as we vary the group parameter (here, angle), and take the tangent vector to this path at each point (derivative with respect to the parameter). We notice that for rotations, any surface which is left invariant under the action also has the property that the tangent vector to the action is tangent to the surface at every point. Another way of stating this is that the tangent vector to the action is normal to the normal vector to the surface. But both of these things are easy to calculate! Suppose the group action is given by a collection of maps $\varphi_t : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. At any point, we can take the tangent vector which is computed by differentiating φ_t with respect to t and setting $t = 0$, so that we get the instantaneous rate of change at that point. We denote this as $V = \left. \frac{d}{dt} \right|_{t=0} \varphi_t$, or more succinctly to save notation, $V = \varphi'_t$. Here, keep in mind that we always plug in $t = 0$ first so that V is a genuine vector at every point and doesn't depend on t . On the other hand, suppose our surface is given by the equation $F = 0$. Then, the normal vector to this surface is given by ∇F . Hence, our condition for invariance is that $V \cdot \nabla F = 0$ whenever $F = 0$ (because we only need the vector to be tangent when looking at points that lie on the surface). Verify that each of these claims make intuitive sense, and check them for the examples we had earlier and come up with other examples.

Exercise (Rotations in $\mathbb{R}^3$)

Suppose we want to rotate $\mathbb{R}^3$ about the origin around the normal vector v . What is the relation of the tangent vector to this action at any point to v ? Note that the tangent vectors also scale linearly with distance from the direction the vector points in. How can you use this to figure out what the tangent vector is at any point (x, y, z) ?

(Come back to this later) Using the tangent vectors, derive the explicit formula for the rotation. Can you come up with a geometric explanation for each part of this formula?

Now we discuss some convenient notation that we use for vector fields. Suppose our vector field has components $V = (f, g, h)$. Instead of writing the functions as components, we instead write it as $V = f\partial_x + g\partial_y + h\partial_z$. We can think of these symbols ∂_{x_i} as just placeholders, but the actual advantage is we can think of V as a differential operator. Now, we can write our condition for invariance as

$$V(F) = (f\partial_x + g\partial_y + h\partial_z)F = f\partial_x F + g\partial_y F + h\partial_z F = 0$$

and this condition $V(F) = 0$ is the same as our dot product condition from earlier, but just written differently. This notation also has other advantages. Usually, to solve for a group action given a tangent vector field, we have to find some set of curves such that the tangent vector at each point is the given vector field. Essentially, if $V = f\partial_x + g\partial_y + h\partial_z$, we must find some set of functions $(X(t), Y(t), Z(t))$ such that $F' = f, G' = g, H' = h$ where the derivatives are taken with respect to t . This process of obtaining the vector field is known as "exponentiation." To explain this, let's discuss what $V(F)$ actually represents. This is a bit of a digression from our main point of Lie groups, but I think it is a useful digression since it gives a unique perspective on some ideas.

There will be some abuse of notation here, but the point is to give the intuition. In particular, dt represents "a small change in the parameter of the group" which we consider as a small amount of time elapsed. If F is any arbitrary function of 3 variables, $V(F)dt = (f\partial_x F + g\partial_y F + h\partial_z F)dt$ is a function which measures the difference $F(x') - F(x)$ where x' is the point after we transform x using our group action over a small period of time dt . This can be seen by noticing $V(F) = (F(x') - F(x))/dt$

which as $dt \rightarrow 0$ is just the definition of the directional derivative (verify this!). Now, let's say that after some finite time τ , we want to compare the value of F at the transformed point and the initial point. Hence, we want to compare $F(g_\tau(x))$ to $F(x)$. If τ is small, the value changes approximately by an amount $F(g_\tau(x)) - F(x) \approx V(F)\tau$, or alternatively, $F(g_\tau(x)) \approx (1 + \tau V)F$ where we consider $V + 1$ as an operator acting on F and 1 is the identity operator where $1F = F$ (we interpret this expression as expanding formally and then applying rules to evaluate things like τVF as derivatives).

Going forward, imagine our vector field is just 1-dimensional acting on $\mathbb{R}$ as $V = d/dx$ (this is the tangent field to the action of translating to the right by t after time t). Think about this action in relation to the function $F(x) = x^2$. Now, this linear approximation is good for small τ , but is bad for larger τ . For example, if we look at $F(x) = x^2$ and want to use $F(1) = 1$ to approximate $F(2)$, the linear approximation gives us $F(2) \approx 3$. But we can do better! What if we decided to do two linear approximations with intervals $1/2$. In other words, we use $F(1)$ to approximate $F(1.5)$ and then use $F(1.5)$ to approximate $F(2)$. We can approximate $F(1.5)$ pretty easily as $F(1.5) \approx 2$, but what do we use as our derivative for our next step? We don't want to use 2 because that would give us the same answer. Instead, we can say that we know the value of our function F in some interval $(1 - \varepsilon, 1 + \varepsilon)$ near one, and use the first order approximation to shift this data to $(1.5 - \varepsilon, 1.5 + \varepsilon)$. In other words, we will take another function $G(x) = (1 + (1/2)\partial_x)F(x)$ which represents "transferring data of F from 1 to 1.5" and approximate $F(2)$ as $(1 + (1/2)\partial_x)G(x)$ which corresponds to "moving this transferred data at 1.5 to 2." In total, we can formally write this as $F(2) \approx (1 + 1/2\partial_x)^2 F(x)$ and evaluate at $x = 1$. You can try this out and see it gives a better approximation. We can keep splitting into smaller and smaller intervals to get formulas $F(2) \approx (1 + (1/n)\partial_x)^n F(x)$ and substitute $x = 1$. You can try this out and see that it gets better and better as we increase n . Essentially this is because we are approximating a far away value by sequentially applying first order approximations, which work very well if we look only at small intervals. We expect then that in the limit we get the exact value, or more suggestively, $F(2) = e^{\partial_x} F(x)$ evaluated at $x = 1$. If we were to translate by a instead, we formally have $F(x + a) = e^{a\partial_x} F(x)$ (go through the logic again and see where the a comes from).

Exercise (Taylor Series as Infinitesimal Exponentiation)

The notation $F(x + a) = e^{a\partial_x} F(x)$ is very suggestive as e^z has a power series expansion. Write out $e^{a\partial_x}$ and act on F with this operator. Change this into a familiar expression that resembles Taylor's theorem. Hence, we have the interpretation that Taylor series can use derivatives to approximate far away values because we're taking a sequence of many small linear approximations.

For a general group action, we get that $F(g_\tau(x)) = e^{\tau V} F$. Here, if we expand things formally, we get terms like $\tau^2 V^2(F)$ which we interpret as $\tau^2 V(V(F))$, and this makes sense as $V(F)$ is a function. It seems like we have just analyzed how to derive the new output of our function given a vector field which seems different than computing the group action itself. However, we can recover the group action from this since the coordinate functions x_i and u are, as the name suggests, functions on the space. Hence, the group action is given by $g_\tau(x) = e^{\tau v} x$. This justifies the name of the exponential of a vector field. If you have studied differential equations before, you will have seen things like the matrix exponential used to solve differential equations. It is essentially the same idea: the differential equations give you a first order infinitesimal rate of change while the exponential compounds these to get a long range, but more complicated, rate of change. If you are familiar with quantum mechanics, there are a lot of places where this idea shows up there as well.

You may have been skeptical throughout this work that sometimes, the transformations yield things which are no longer valid functions. For example, you can rotate a line to become a vertical line, but this is no longer a function. The solution to this is to consider "local group actions" which essentially states for example that if you rotate a line too much, it may fail to be a function, but if you fix a line, you can always restrict your angle sufficiently small so that if we rotate within this range, the resulting graph will still be a valid function. The idea is instead of looking at the full group $\mathbb{R}^1$ or S^1 acting on the space, we only look at a neighborhood of the identity acting on the space (rotate by a sufficiently small angle). This is the sense in which we can consider the actions of $\mathbb{R}^1$ and S^1 to

be the same even though the groups themselves are different; near the identity, both groups look like an interval, and hence are essentially the same. This viewpoint is perfectly fine to test for invariance since what we have just glimpsed into is that we can test invariance with derivatives, which are a local property. In the next section, we will see how to use this method to understand the group action on differential equations. A more detailed exposition of the local group action can be found in the book.

2.2 Vector Field Prolongation

So far, we've seen how to check if a group action left a differential equation invariant by understanding the prolonged action on the jet space. We realized that the method for checking the invariance was quite inefficient, and found an alternative geometric criterion to determine whether or not a hypersurface is left invariant under a group action. We found that an infinitesimal criterion was useful, namely if V is the tangent vector field to our group action, we can determine whether a hypersurface is left invariant by the condition that $V(F) = 0$. We will see how to connect these two languages of the jet space and the infinitesimal criterion for invariance to find a nice method of finding out whether a group leaves a differential equation invariant.

Here's how the method will proceed: we differentiate the action of our group on our base space to get the tangent vector field. We can also prolong our group action and test for if this leaves the differential equation invariant by using the infinitesimal criterion instead of checking algebraically. The tangent field of this prolonged action in the jet space will be called the prolonged vector field. It turns out in fact that we can simplify our lives greatly by finding a formula for this prolonged vector field solely in terms of the original vector field and its derivatives! The advent of this is we will avoid the issues we found last time of having to choose an arbitrary function and computing the inverse of the chosen function; we bypass the function altogether! This is possible because whether an equation is invariant under an action is solely a local property, and the prolonged vector field allows us to do this computation using no more than local information. At last, we have arrived at the motivation for why we care about the continuity of the group (action): we want to be able to differentiate it to get a very easy criterion of invariance. This technique is very powerful and can actually be used to determine all possible group actions that leave a group invariant. We will use this to analyze the heat equation and get some very surprising results.

It turns out that we can determine the equation for the vector field prolongation without ever referencing an intermediate function. We denote the original vector field as $v = \sum \xi_i \partial_{x_i} + \phi \partial_u$. Here, ξ_i and ϕ is a function of x_i, u as the values depend on the point in the base space. Then, the prolonged vector field of the group action on $U^{(k)}$ is written as $pr^{(k)}v = v + \sum_{|J| \leq k} \phi^J \partial_{u_J}$ (why is it v plus something?). Here, the superscripts on ϕ is just a label to associate them with the corresponding derivative of u . The way to think of the functions ξ_i, ϕ, ϕ^J is that they measure how much the corresponding coordinate changes with a small application of the group action. For example, if we apply the group for a short period of time dt , we will get (primed variables represent transformed variables/values) $x'_i - x_i \approx \xi_i dt$, $u' - u \approx \phi dt$ and $u'_J - u_J \approx \phi^J$. This is just saying the derivative is the best first order approximation.

Then, the theorem states that we have the explicit formula

$$\phi^J = D_J \left(\phi - \sum \xi_i u_{x_i} \right) + \sum \xi_i u_{J x_i}$$

Here, D_J represents the "total derivative" of a variable with respect to the multi-index J . We think of each variable not as just an independent variable, but also as a function depending on the variable we're differentiating with respect to. So the "usual derivative" of u with respect to x is 0 because they're both just coordinate variables in the jet space, but $D_x(u) = u_x$. Also, $u_{J x_i}$ is first taking the derivative of u with respect to J and then with respect to x_i .

These expressions can get very messy. For example, if ξ depends on x, u , we have that $D_x(\xi) = \xi_x + \xi_u u_x$. If we have a multi-index like $D_{xt}(\phi)$, we interpret this as $D_x(D_t(\phi))$. We will see that the part inside the D_J is the main contribution and the part we add on at the end is just to cancel out some

higher order derivatives that show up. For example, when we look at how much the first derivative changes, the D_J term involves something involving a second derivative, which shouldn't affect our answer. The last term serves to cancel this problematic bit.

The proof is some technical calculation, and is outlined in the book. We will only be using it as a computational tool and so the only important thing to understand is that such a formula exists which can calculate the prolonged vector field using only the components of the original vector field, and not relying on another function. We can discuss some of the ideas of how we use locality to bypass the function. The proof proceeds by proving it for $U^{(1)}$, then using induction seeing that $U^{(n)}$ can be embedded into $V^{(1)}$ where $V = U^{(n-1)}$ (n -th derivatives can be considered as first derivatives of $n-1$ -th derivatives). We first use the fact that the transformed function of $u = f(x)$ is $\tilde{u} = g_t f(g_t^{-1} \tilde{x})$. Now, we want to look locally so we take the Jacobian of the transformation $g_t : U^{(0)} \rightarrow U^{(0)}$ to get a linear map G_t which essentially encodes the local data and gets rid of the dependence of the function itself (we only look at the first derivative of it). Now, we differentiate this linear map with respect to t and set $t = 0$ to obtain the tangent vector field.

3 Lecture III

3.1 Revisiting $u_{xx} = 0$

Let's go back to our motivating example and show that rotation will leave this space invariant. Our action is $g_t(x, u) = (x \cos t - u \sin t, u \cos t + x \sin t)$, and the tangent vector to this action on $U^{(0)}$ is $\frac{d}{dt}|_{t=0} g_t(x, u) = -u\partial_x + x\partial_u$. Now, we can find what ϕ^x (how much the first derivative changes) in terms of our explicit formula. Here, $J = x, \xi = -u, \phi = x$. Plugging things into our equation, we have

$$\phi^x = D_J(\phi - \xi u_x) + \xi u_{Jx} = D_x(x + uu_x) - uu_{xx} = 1 + u_x^2 + uu_{xx} - uu_{xx} = 1 + u_x^2.$$

Notice that the final term was there to cancel out the second derivative term which we expect shouldn't be there, as was explained earlier. Now, how do we determine how our first derivative actually changes? Well, ϕ^x represents how much u_x changes to the first order in the sense that $\frac{\partial u_x}{\partial t} = 1 + u_x^2$. We also need to supply initial data which is our initial value of our first derivative. We denote this by γ . Now we solve this equation

$$\frac{\partial u_x}{\partial t} = 1 + u_x^2 \implies \arctan(u_x) = t + C \implies u_x = \tan(t + C)$$

$$\gamma = u_x(0) = \tan(C) \implies u_x = \tan(t + C) = \frac{\tan(t) + \tan(C)}{1 - \tan(t)\tan(C)} = \frac{\tan(t) + \gamma}{1 - \gamma \tan(t)} = \frac{\cos(t) + \gamma}{1 - \gamma \sin(t)}$$

which is precisely the formula we obtained by the prolonged group action, except we didn't need to choose a function! If you aren't convinced this method makes our lives significantly easier, let's compute the action on the second derivative. Using our formula again, we want to compute ϕ^{xx} . Here, $J = (x, x)$ so we have

$$\phi^{xx} = D_{xx}(x + uu_x) - uu_{xxx} = D_x(1 + u_x^2 + uu_{xx}) - uu_{xxx} = 2u_x u_{xx} + u_x u_{xx} + uu_{xxx} - uu_{xxx} = 3u_x u_{xx}$$

where again the final term cancels out the problematic derivative that has an order too high. Now, we want to go and solve $\frac{\partial u_{xx}}{\partial t} = 3u_x u_{xx}$ with initial data $\gamma = u_x(0)$ and $\delta = u_{xx}(0)$. Now, we have two different independent variables in our differential equation which both depend on t (since the group action simultaneously changes the first and second derivatives). In order to remove the dependence on u_x , we can just apply the formula we just derived! We can substitute $u_x = \tan(t + c)$ (with $\tan(c) = \gamma$) and solve

$$\frac{\partial u_{xx}}{\partial t} = 3 \tan(t + c) u_{xx} \implies \log(u_{xx}) = -3 \log(\cos(t + c)) + c' \implies u_{xx} = C' (\cos(t + c))^{-3}$$

I leave it to you to verify that when we use the information $\delta = u_{xx}(0) = C'(\cos(c))^{-3}$ and $\tan(c) = \gamma$ to show that the solution we get is $u_{xx} = \frac{\delta}{\cos(t) - \gamma \sin(t)}$. This is in fact the same equation that we got earlier, and in particular it shows that rotations send points on a graph with zero second derivative to other points with that property.

In fact, we didn't even have to compute our prolonged group action. We could have used our infinitesimal invariance criterion to see that rotations preserve $u_{xx} = 0$. Remember that if we have a vector field of a group action V and want to check that it leaves a hypersurface $F = 0$ invariant, we just have to check $V(F) = 0$ whenever $F = 0$. In this case, $V = pr^{(2)}v$ and $F = u_{xx}$ so we just need to check that $pr^{(2)}u_{xx} = 0$ when $u_{xx} = 0$. Here, we get $pr^{(2)}u_{xx} = (-u\partial_x + x\partial_u + (1+u_x^2)\partial_{u_x} + 3u_x u_{xx}\partial_{u_{xx}})u_{xx} = 3u_x u_{xx} = 0$ since $u_{xx} = 0$. This is the most direct way to get that the differential equation is left invariant under the group action!

3.2 Heat Equation

Our new language of prolonged vector fields can actually do more than just verify that a group action leaves a differential equation invariant. In fact, it allows us to determine all possible group actions that leave a differential equation invariant! We will explore this with the specific example of the heat equation $u_t = u_{xx}$. To outline the procedure, we will leave the components of our base space vector field as variables depending on x_i, u . Then, we will compute the prolongation of this vector field up to the degree of the highest degree derivative in the differential equation (in this case, 2). Letting this prolonged field act on our differential equation of interest, we get differential equations on the components of the base space vector field. Solving these, we can obtain all possible symmetries of our desired equation.

In this case, our base space vector field is of the form $v = \xi\partial_x + \tau\partial_t + \phi\partial_u$ where ξ, τ, ϕ are all functions of x_i, u . Since we go up to the second derivative, we must compute up to the second prolongation which looks like

$$pr^{(2)}v = \xi\partial_x + \tau\partial_t + \phi\partial_u + \phi^x\partial_{u_x} + \phi^t\partial_{u_t} + \phi^{xx}\partial_{u_{xx}} + \phi^{xt}\partial_{u_{xt}} + \phi^{tt}\partial_{u_{tt}}.$$

If we let this vector field act on our differential equation $u_{xx} - u_t = 0$ and invoke our invariance condition, we obtain that $\phi^{xx} - \phi^t = 0$. In other words, $\phi^t = \phi^{xx}$ when $u_t = u_{xx}$. To find all admissible vector fields, we must first solve for ϕ^t and ϕ^{xx} using our prolongation formula. We have

$$\phi^t = D_t(\phi - \xi u_x - \tau u_t) + \xi u_{xt} + \tau u_{tt} = \phi_t - \xi_t u_x + (\phi_u - \tau_t)u_t - \xi_u u_x u_t - \tau_u u_{xx}^2$$

and

$$\begin{aligned} \phi^{xx} &= D_{xx}(\phi - \xi u_x - \tau u_t) + \xi u_{xxx} + \tau u_{xxt} = \phi_{xx} + (2\phi_{xu} - \xi_{xx})u_x - \tau_{xx}u_t + (\phi_{uu} - 2\xi_{xu})u_x^2 \\ &\quad - 2\tau_{xu}u_x u_t - \xi_{uu}u_x^3 - \tau_{uu}u_x^2 u_t + (\phi_u - 2\xi_x)u_{xx} - 2\tau_x u_{xt} - 3\xi_u u_x u_{xx} - \tau_u u_t^2 - 2\tau_{uu}u_x u_{xt}. \end{aligned}$$

The next step is to get rid of worrying about the condition of when $u_t = u_{xx}$. We do this simply by replacing anywhere we see u_t with u_{xx} in the above equations. Then, since x_i, u, u_J are all coordinate functions in our jet space, they can take arbitrary values. By a standard argument, $\phi^t = \phi^{xx}$ if and only if the coefficients of each monomial in x_i, u, u_J match on both sides (after making the $u_t = u_{xx}$ substitution). For example, if we look at the coefficient of $u_x u_{xt}$ we get that $0 = -2\tau_u$ since the coefficient is 0 for ϕ^t and $-2\tau_u$ in ϕ^{xx} . This condition tells us that τ is independent of u . If further we substitute $u_t = u_{xx}$ and look at the resulting coefficients of $u_x u_{xx}$, we get $-\xi_u = -2\tau_{xu} - 3\xi_u$ (verify). But we just determined $\tau_u = 0$, so this tells us that $\xi_u = 0$ or that ξ is also independent of u . The equations are messy, but the upshot is that it gives us a lot of information. Continuing in this manner (you can try it as an exercise, or look at the details in the book), we get all possible admissible vector fields are of the form

$$\begin{aligned} \xi &= c_1 + c_4 x + 2c_5 t + 4c_6 x t \\ \tau &= c_2 + 2c_4 t + 4c_6 t^2 \\ \phi &= (c_3 - c_5 x - 2c_6 t - c_6 x^2) + \alpha(x, t) \end{aligned}$$

where c_i are arbitrary constants and α is any arbitrary function that satisfies $\alpha_t = \alpha_{xx}$, or in other words, is itself a solution to the heat equation. We have then that all admissible vector fields of group actions that leave the heat equation invariant are of the form $v = \xi\partial_x + \tau\partial_t + \phi\partial_u$ where ξ, τ, ϕ are of the form prescribed above. We can set all c_i except one equal to 0 to get the following set of "basis vector fields":

$$\begin{aligned} c_1 : v_1 &= \partial_x \\ c_2 : v_2 &= \partial_t \\ c_3 : v_3 &= u\partial_u \\ c_4 : v_4 &= x\partial_x + 2t\partial_t \\ c_5 : v_5 &= 2t\partial_x - xu\partial_u \\ c_6 : v_6 &= 4tx\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u \end{aligned}$$

and an infinite class of vector fields

$$v_\alpha = \alpha(x, t)\partial_u.$$

We have that any vector field can be written as $v = \sum c_i v_i + v_\alpha$ for suitable c_i, α . The full group of symmetries being a group ends up implying that $\exp(av)$ can be written as a (group) product of $\exp(a_i v_i)$ and $\exp(a_\alpha v_\alpha)$ for suitable a_i, a_α . This means that we can study $\exp(av_i), \exp(av_\alpha)$ and taking the group generated by these actions will give us all possible actions. Let's list these out. We shall write G_i as the action corresponding to v_i , and we will write what the image of (x, t, u) is after transforming with parameter ε

$$\begin{aligned} G_1 : (x + \varepsilon, t, u) \\ G_2 : (x, t + \varepsilon, u) \\ G_3 : (x, t, e^\varepsilon u) \\ G_4 : (e^\varepsilon x, e^{2\varepsilon} t, u) \\ G_5 : (x + 2\varepsilon t, t, u \exp(-\varepsilon x - \varepsilon^2 t)) \\ G_6 : \left(\frac{x}{1 - 4\varepsilon t}, \frac{t}{1 - 4\varepsilon t}, u \sqrt{1 - 4\varepsilon t} \exp\left(\frac{-\varepsilon x^2}{1 - 4\varepsilon t}\right) \right) \\ G_\alpha : (x, t, u + \varepsilon \alpha(x, t)). \end{aligned}$$

You can check that each of these is the group action corresponding to the tangent vector field above. Let's comment on some of these. The first two are translations in the independent variable, and this leaves our differential equation invariant because every term involves a derivative, and derivatives don't change by translating. The third action along with G_α account for linearity since α is another solution of the differential equation. It is interesting to see that this method can capture even the linearity of a differential equation, since even though it is probably the first invariance you learn, it seems much more complicated here than say, detecting translation or scaling invariance. The fourth and fifth are also interesting, but I hope that you will agree with me to say that the sixth transformation is truly something bizarre and could not have possibly been guessed without this systematic method. This demonstrates the power of utilizing this powerful machinery.

As a final remark on this problem, if you take the solution $u = c$ for some constant, apply G_6 , and follow by an appropriate translation in t , you will find that $u = \frac{1}{\sqrt{4\pi t}} \exp\left(\frac{-x^2}{4t}\right)$ is a solution. In other words, this method was able to start from an almost trivial solution of the heat equation and deduce that the gaussian is also a solution! This is truly remarkable.

You can apply this method in many many more contexts outlined in the book. You can, for example, account for multiple independent variables and study the symmetries of the Navier-Stokes equations. You can also analyze the wave equation and find some classical symmetries such as translations and Gallilean boosts, but you can also find that sphere inversion is a symmetry of the equation!

An application later in the book is to use similar ideas to study Noether's theorem, a powerful statement in physics which relates invariant quantities like momentum and energy, to the invariance of a differential equation under certain transformations in the conjugate variables, in this case position and time respectively. This is truly a beautiful subject, and hopefully these notes have motivated you to look into this application of Lie groups more deeply, or at the very least, be convinced that these objects are interesting to study.

Exercise ($u_{xx} = 0$ One Last Time)

Use the same method as above to compute all symmetries of the differential equation $u_{xx} = 0$. Show that the most general group action leaving this transformation invariant is the projective group of the plane (eight dimensional). Geometrically, we act on lines by the projective group by considering the projective duals. So in a sense, these are all geometrically all possible actions which send lines to lines. We have thus obtained a nice geometric fact by analyzing the invariance of a differential equation!

Acknowledgements

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